

Anchoring Large Language Models to Empirical Behavioral Evidence

A Hybrid Approach for DCE-Grounded Narrative Policy Simulation

Abstract

Objective: To develop empirical-frontier regularization (EFR), an inference-time procedure anchoring LLM outputs to a discrete choice experiment (DCE) reference surface.

Methods: A Wuhan vaccination DCE ($N = 1,027$) was combined with LLM simulation on a 64-state policy grid. EFR computes $\pi_{\text{EFR}} = \lambda \pi_{\text{LLM}} + (1 - \lambda) P_{\text{emp}}$ with $\lambda = 0.25$.

Results: Independent local replication validation on six administered DCE tasks ($N = 205$ respondents) showed multinomial log loss: Pure-DCE 0.911, EFR 0.923, Raw LLM 1.020, Uniform 1.099. EFR improved over Raw LLM (9.5% log loss reduction) and approached Pure-DCE (within 1.3%). On a 64-state grid, waiting-time monotonicity violations fell from 10.4–62.5% to 7.3–30.2% across three LLMs under EFR.

Limitations: All respondents completed identical tasks; validation evaluates generalization to unseen respondents, not unseen tasks. Out-of-design scenarios lacked human ground truth.

Conclusion: EFR is a constrained narrative-extension mechanism for an existing DCE, not a replacement for the underlying model or validation of LLMs as standalone behavioral simulators.

Keywords: empirical-frontier regularization; large language models; discrete choice experiment; health technology assessment

Article Type: Methodological Article

Highlights (120 words maximum)

1. EFR is an inference-time regularization layer that constrains LLM outputs to a DCE-estimated reference without retraining.
2. On held-out respondent validation, EFR recovers most of the DCE model's predictive performance while retaining a bounded LLM component.
3. EFR improves structural consistency (monotonicity, rank order) but does not establish behavioral validity for out-of-design narrative scenarios.

1 Introduction

Large language models (LLMs) are increasingly used as simulated decision-makers for healthcare policy. Their appeal is interpretability of rich natural-language scenarios that structured models cannot directly process. However, linguistic fluency does not guarantee behavioral reliability: unconstrained LLMs may violate basic preference relationships implied by observed human behavior.

We propose empirical-frontier regularization (EFR), an inference-time procedure anchoring LLM outputs to a DCE-estimated reference surface without retraining. The method blends raw LLM probabilities with an empirical choice frontier through a transparent weighted average ($\lambda = 0.25$ fixed). The adjustment is fully auditable: raw LLM probability, empirical frontier value, blending weight, and final output are directly inspectable.

2 Methods

2.1 Empirical Frontier

A grouped conditional logit was estimated from a Wuhan vaccination-timing DCE ($N = 1,027$ adults, fielded December 2022–January 2023). The frontier defines $P_{\text{emp}}(x)$, the DCE-implied binary policy-versus-opt-out probability for any attribute profile x .

2.2 EFR Procedure

For each policy scenario, EFR computes:

$$\pi_{\text{EFR}} = \lambda \cdot \pi_{\text{LLM}} + (1 - \lambda) \cdot P_{\text{emp}}$$

with $\lambda = 0.25$ as an illustrative conservative operating point. The empirical component P_{emp} supplies the behavioral signal; the LLM component is permitted only inside this convex blend.

2.3 Evaluation

Three open-source LLMs were evaluated: Qwen2.5-72B-Instruct, DeepSeek V4, and MiroThinker-1.7-mini. Two validation domains were used:

(1) **Structural consistency:** A 64-state grid spanning DCE attribute dimensions. Metrics: waiting-time monotonicity violation rate (MVR-Wait), rank correlation with P_{emp} .

(2) **Held-out predictive:** Six administered DCE choice tasks from 205 held-out respondents (20% split). Metric: multinomial log loss against empirical choices.

3 Results

3.1 Held-Out Predictive Validation

Independent local replication validation (local Ollama Qwen2.5-72B) showed multinomial log loss: Pure-DCE 0.911, EFR 0.923, Raw LLM 1.020, Uniform 1.099. EFR materially improved over Raw LLM (9.5% reduction) and approached Pure-DCE (within 1.3%).

3.2 Structural Consistency

On the 64-state grid, MVR-Wait fell from 10.4–62.5% to 7.3–30.2% across three LLMs under EFR anchoring (Table 1). Frontier-alignment rank correlation rose from 0.061–0.531 (Raw LLM) to 0.689–0.812 (EFR).

4 Discussion

EFR is best understood as a constrained narrative-extension mechanism for an existing DCE. The empirical reference, not the LLM, supplies the behavioral signal. Whether narrative extension improves behavioral accuracy on out-of-design scenarios remains to be established by future work with human ground truth.

Limitations. The Wuhan DCE is a single-jurisdiction sample. The respondent-level holdout evaluates generalization to unseen respondents, not unseen choice tasks. Out-of-design narrative scenarios lacked human ground truth.

Implications. With a DCE, EFR provides a path from evidence to narrative policy simulation. Without a DCE, the framework is not a substitute for stated-preference data. Best suited for early-stage screening, not high-stakes final decisions.

References

References

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